

Paper ED-Cos-01 — Inflation as ED Diffusion Outpacing ED Production

A Generative Paper of the Event Density Program

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Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of quantitative inflation parameters ($N \sim 60$, $n_s \approx 0.965$, r). All INHERITED from Planck CMB; substrate-graph derivation OPEN.
2. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), not the operational-reconstruction lineage.
3. It does **not** posit an inflaton field, potential $V(\phi)$, slow-roll parameters, or reheating dynamics. These GR machineries are absent at substrate level; the substrate-side **proposed** replacement (diffusion-production imbalance + saddle-Hessian dynamics) is identified at vocabulary level — substrate-graph qualitative-mechanism derivation OPEN.
4. It does **not** claim full qualitative-mechanism closure of all substrate-graph derivations. **Status update (2026-05-16):** (a) substrate-graph derivation of exponential scale-growth $\dot{a} \propto a$ **CLOSED at D-via-I (approximately-vacuum-energy level)** via the audited M3 substrate-graph chain (§3.8); (b) regime duration ~ 60 e-folds equivalent — substrate-graph derivation still OPEN; (c) smooth termination — substrate-graph derivation still OPEN; (d) V1-transported substrate-graph noise producing near-power-law spectral character — now partially identifiable via small deviations from $w = -1$ in non-uniform realistic regime; quantitative spectrum still OPEN; (e) inflation's contribution to horizon resolution beyond SCBU inheritance — now identifiable via M3 chain producing exponential extension; substrate-graph quantitative comparison with SCBU contribution still OPEN.
5. **Regime caveat:** the substrate-side identification inherits DCGT's A $\rightarrow$ regime hydrodynamic-window scale-separation assumption ($\ell_{ED} \ll R_{cg} \ll L_{flow}$) per Paper_073 + ED_MEMORY anchor. **"No fine-tuning"** is not claimed — the fine-tuning question is **relocated** from $V(\phi)$ shape + initial conditions to substrate-graph parameters of the $\Gamma_{diff}/\Gamma_{prod}$ regime; whether the relocation is a genuine improvement requires substrate-graph closure (OPEN). No new substrate primitives.

Abstract

Cosmic inflation in ED is the transient post-SCBU substrate sub-regime where diffusion-rate exceeds production-rate, producing structural transition from the boundary-scale-collapsed state (Paper_ED_CCC §3.6) to the standard individuation regime (Paper_072). Substrate diffusion (Paper_073 DCGT under V1 + V5 finite-width content) and substrate production (Paper_012 P-RB-1 commitment-event rate) admit a regime in the post-SCBU ignition phase (Paper_ED_CCC §3.7) where $\Gamma_{diff} > \Gamma_{prod}$. Under this imbalance, the substrate-action saddle's (Paper_ED_SC_4_9) compression-dominant Hessian axes deplete; expansion-dominant axes

are sustained by diffusion-driven redistribution at approximately-constant substrate-action density. **DCGT translates this constant density to a vacuum-energy-like continuum stress-energy with $w \approx -1$; Friedmann recovery gives constant Hubble parameter H and exponential scale-growth $a(t) = a_0 e^{Ht}$** (M3 substrate-graph chain Steps A–E; §3.8). The substrate-c bound is preserved at substrate-graph level; exponential growth is a continuum-side coarse-grained-geometry effect via the same de-Sitter mechanism standard cosmology uses. Horizon-problem resolution receives both SCBU inheritance (Paper_ED_CCC §3.6) and an inflation-side contribution (audit row 12 partially closed). Inflaton-particle ontology absent. **Verdict per Paper_095: M3 (form-IDENTIFIED + value-INHERITED) within DCGT A→regime** — upgraded from M2 following audited closure of load-bearing row 13 via M3 chain. **F1 (sharpest):** detection of an inflaton-particle quantum refutes the substrate-ontology framing.

1 Statement of Result

Cosmic inflation in ED is **proposed** as the structural composition of four substrate elements:

1. **Substrate diffusion content** (Paper_073 DCGT): V1 + V5 finite-width kernels supply diffusion-rate Γ_{diff} in the hydrodynamic window.
2. **Substrate production content** (Paper_012 P-RB-1): P11 commitment events occur at the substrate rate-of-becoming, populating substrate with individuated content (Paper_072) at rate Γ_{prod} .
3. **Saddle-Hessian gradient-sign partition** (Paper_ED_SC_4_9): $S_{\text{sub}}[\Psi]$ partitions each locus into compression-dominant and expansion-dominant orthogonal Hessian axes.
4. **Post-SCBU ignition regime** (Paper_ED_CCC §3.7): substrate transitions from boundary-scale-collapsed (SC-4.x) state into populated-individuated state; the two rates are non-stationary across this transition.

Inflation = proposed transient sub-regime where $\Gamma_{\text{diff}} > \Gamma_{\text{prod}}$, producing an unbalanced Hessian-signature pattern (expansion-dominant axes sustained; compression-dominant depleted). The Hessian-signature pattern propagates outward via V1 at substrate-c, growing the substrate-graph region within which the unbalanced-saddle structure obtains. Inflation ends when Γ_{prod} recovers; saddle re-equilibrates; standard individuation resumes.

Primary horizon-resolution: SCBU inheritance (Paper_ED_CCC §3.6 + SC-4.x scale-collapse), not inflation. Inflation supplies the **structural transition dynamics** from boundary state to standard individuation; whether it additionally amplifies horizon resolution beyond SCBU inheritance is OPEN (audit row 12).

Form-IDENTIFIED at substrate-graph level via the audited M3 chain (§3.8); value-INHERITED (N, n_s, r from Planck CMB). **Verdict: M3 (form-IDENTIFIED + value-INHERITED) within DCGT A→regime per §4 audit table** (upgraded from M2 following row 13 closure).

2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02, P04, P07, P09, P11, P12, P13.

Upstream: Paper_012 (P-RB-1 substrate-c, commitment-event rate); Paper_072 (individuation regime); Paper_073 (DCGT substrate diffusion content, A→regime); Paper_089 (V1 retarded kernel); Paper_090 (V5 cross-chain); Paper_093 (T18 kernel-arrow); Paper_ED_SC_4_9 (substrate-action saddle Hessian); Papers SC-4.x (scale-collapse); Paper_ED_CCC §3.6 + §3.7 (post-SCBU homogeneity + ignition); Paper_ED_GW_00 (saddle-Hessian reconfiguration via V1); Paper_ED_Baryogenesis (M2 verdict-framing template); Planck CMB (INHERITED).

Paper-specific naming convention (no new postulate):

- **P-Inflation-Identification:** Naming convention for the proposed substrate-side identification “*Inflation = transient post-SCBU ignition sub-regime where $\Gamma_{\text{diff}} > \Gamma_{\text{prod}}$, producing outward V1 propagation of an unbalanced saddle-Hessian-signature pattern.*” Definitional per Paper_095 §2.3; vocabulary-level identification only; substrate-graph qualitative-mechanism closure OPEN.

No new primitives. No paper-specific postulates beyond the naming convention.

3 Derivation

3.1 Ontology: saddle geometry, no metric, no fields

Per Paper_087, no fundamental spacetime metric, no fundamental scalar field, no fundamental potential, no Hilbert-space mode. Per Paper_ED_SC_4_9, $S_{\text{sub}}[\Psi]$ admits saddles partitioning each locus into compression-dominant and expansion-dominant Hessian axes. Inflation, like GW phenomenology (Paper_ED_GW_00), is **dynamics-class** — saddles in motion — not kinematics on a fixed background.

3.2 Substrate diffusion and production rates

Per Paper_073 DCGT (within hydrodynamic window $\ell_{ED} \ll R_{cg} \ll L_{flow}$), V1 + V5 supply diffusion-rate $\Gamma_{\text{diff}}(\ell)$ as a substrate-graph quantity. Per Paper_012 P-RB-1 + P11, production-rate $\Gamma_{\text{prod}}(\ell)$ is the rate at which commitment populates substrate with individuated content (Paper_072). Three regimes: production-dominated (standard); balanced (equilibrium-like); diffusion-dominated (the proposed inflation regime).

3.3 The diffusion–production imbalance

When $\Gamma_{\text{diff}} > \Gamma_{\text{prod}}$, diffusion redistributes Ψ content faster than commitment can replenish. Compression-dominant Hessian axes deplete; expansion-dominant axes are sustained by diffusion-driven redistribution. Saddle-Hessian gradient-sign structure is temporarily unbalanced in favor of expansion. **Whether the substrate naturally produces this regime in the post-SCBU ignition phase is structurally plausible from Paper_ED_CCC §3.7 non-stationarity** but not constructively derived; substrate-graph closure OPEN (audit row 11).

3.4 Outward V1 propagation of the unbalanced-saddle pattern

The substrate-graph object that propagates is the **unbalanced-saddle Hessian-signature pattern**, not amplitude in a fixed Ψ -direction nor a spatial region per se. V1 retarded propagation along the kernel-arrow (Paper_089 + Paper_093 T18) carries this pattern outward at substrate-c — the substrate-graph region within which the unbalanced-saddle structure obtains grows at substrate-c.

Direct substrate-c-bounded region-growth gives at most LINEAR scale-growth $a(t) \sim t$ at substrate-graph level. **Exponential scale-growth** $a(t) = a_0 e^{Ht}$ **characteristic of standard inflation now closes via the M3 substrate-graph chain (§3.8 below):** the substrate-c bound is preserved at substrate-graph level, and exponential growth emerges as a continuum-side coarse-grained-geometry effect via DCGT translating the saturation-regime constant substrate-action density to a vacuum-energy-like continuum stress-energy. Same de-Sitter mechanism standard cosmology uses (bounded local light-speed, exponential global scale-growth via positive vacuum-energy curvature).

The GR-side Hubble parameter H acquires a substrate-graph identification via $H = \sqrt{(8\pi G_{\text{eff}}/3)\rho_{\text{eff}}^{\text{const}}}$, where $\rho_{\text{eff}}^{\text{const}}$ is the DCGT-translated saturation-regime substrate-action density. The quantitative value derivation is OPEN-4 (still OPEN; quantitative consolidation, not load-bearing for M3 framing).

3.5 Termination — production catches up

Inflation ends when Γ_{prod} recovers. Two structural mechanisms supplied by the post-SCBU ignition phase (Paper_ED_CCC §3.7): event-density climbs through ignition, lifting commitment frequency per Paper_012; expansion-dominant axes extending over larger substrate-graph regions reduce per-locus diffusion-rate. When $\Gamma_{\text{prod}} \gtrsim \Gamma_{\text{diff}}$, gradient-sign structure re-equilibrates; standard individuation regime (Paper_072) resumes. **Whether termination is smooth (not abrupt) and whether the regime duration corresponds to GR-side ~ 60 e-folds requires substrate-graph closure — OPEN** (audit rows 14, 15). The mechanisms above make smooth, finite-duration termination structurally plausible but not derived.

3.6 Horizon problem: primarily SCBU inheritance

Observable	Substrate-side identification	Primary source
Horizon problem	Cosmic-scale homogeneity at SCBU boundary leaves the post-boundary substrate already in causal contact; inflation’s role is structural transition into standard individuation	SCBU inheritance (Paper_ED_CCC §3.6 + SC-4.x). Inflation’s non-redundant contribution OPEN (row 12).
Flatness	SC-4.x scale-collapse removes local curvature content at the boundary; post-boundary substrate inherits flatness	SCBU inheritance (Paper_ED_CCC §3.6 + Papers SC-4.x)
Isotropy	Globally-coherent kernel-arrow (Paper_ED_CCC §3.7); V1 propagates isotropically	SCBU inheritance
Primordial fluctuations	Substrate-graph noise in Ψ during diffusion-dominated phase, transported by V1	Identified at vocabulary level; qualitative power-law character OPEN (row 16); quantitative spectrum INHERITED from Planck

The honest framing: **homogeneity, flatness, isotropy resolve via the SCBU boundary** (already in the corpus). Inflation supplies the post-boundary dynamics, not the homogeneity-establishing mechanism. This avoids double-counting and matches Paper_ED_CCC’s existing position.

3.7 Contrast with GR-side scalar-field inflation

Element	GR-side	ED substrate-side	Status
Driver	Inflaton field ϕ on potential $V(\phi)$	$\Gamma_{\text{diff}} > \Gamma_{\text{prod}}$ regime	Identified
Mechanism	Slow-roll on flat potential	Saddle-Hessian imbalance + V1 pattern propagation	Identified

Element	GR-side	ED substrate-side	Status
Termination	Inflaton at potential minimum; reheating	Production-rate recovery; saddle re-equilibration	Identified
Expansion rate	$H = \sqrt{V/3M_{Pl}^2}$	Substrate-c-bounded region-growth via DCGT vacuum-energy translation	D-via-I (M3 chain)
Fine-tuning	$V(\phi)$ shape, initial conditions, slow-roll	Relocated to substrate-graph regime parameters	Identified (not solved)
Particle content	Hypothetical inflaton particle	None at substrate level	Replaced
Phenomenology	Horizon resolution, flatness, isotropy, fluctuations	Same observables; primary horizon-resolution via SCBU	Inherited + Identified

With the M3 chain closure, the Driver / Mechanism / Termination / Expansion-rate entries are now D-via-I-substrate-graph-derivable (at approximately-vacuum-energy / approximately-constant levels per audited closures), not just vocabulary-Identified.

4 M3 Substrate-Graph Chain Summary

The audited substrate-graph closure for exponential scale-growth $\dot{a} \propto a$ proceeds via a five-step chain established across seven exponential-growth memos. Steps A, D, E are standard inheritance; Steps B, C are D-via-I closures audit-accepted at approximate-level qualifications matching standard cosmology slow-roll inflation discipline.

Step	Substance	Status	Reference
A	Saturation regime structure: $\Gamma_{\text{diff}} > \Gamma_{\text{prod}}$ in post-SCBU ignition phase produces unbalanced saddle-Hessian content (expansion-dominant sustained, compression-dominant depleted)	INHERITED	§3.3; Paper_ED_CCC §3.7
B	Substrate-action density $s(\ell, t) = S_{\text{sub}}/V$ approximately constant across saturation region (spatially via SC-4.x homogeneity + uniform channel populations; temporally via dynamic-equilibrium saturation-condition + boundary-locus density inheritance)	D-via-I CEPTED approximately-constant level)	(AC- at + Audit Memo_ED_SubstrateAction_
C	DCGT translates approximately-constant substrate-action density to approximately-constant vacuum-energy-like continuum stress-energy via substrate-side Noether (P03 + P13 substrate-translation invariance; standard lattice-field-theory analog) $\rightarrow T_{\text{eff}}^{\mu\nu} \approx -g_{\text{eff}}^{\mu\nu} \rho_{\text{eff}}^{\text{const}}$ with equation of state $w \approx -1$	D-via-I CEPTED approximately-vacuum-energy level)	(AC- at + Audit Memo_ED_DCGT_VacuumE
D	Friedmann recovery from vacuum-energy stress-energy: $H^2 = (8\pi G_{\text{eff}}/3)\rho_{\text{eff}}^{\text{const}} = \text{constant}$	INHERITED	Standard FRW + Friedmann
E	$a(t) = a_0 e^{Ht}$ — exponential scale-growth	INHERITED	Standard de-Sitter solution

Key structural feature: the substrate-c bound is preserved at substrate-graph level throughout.

Exponential growth is a continuum-side coarse-grained-geometry effect via DCGT — same mechanism standard cosmology uses (bounded local light-speed produces exponential global scale-growth via positive vacuum-energy curvature; cosmological-constant de Sitter).

Approximation-level qualifications (per audited closures):

- Strict $w = -1$ for uniform-saturation; small deviations for non-uniform realistic ignition regime (consistent with standard slow-roll inflation phenomenology — substrate-side source of primordial fluctuation spectrum’s slight red tilt).
- Bulk-of-regime, not start/end transitions or sub-coarse-graining-scale resolution.
- Discrete-Noether substrate-graph derivation INHERITED from standard lattice-field-theory; explicit substrate-graph construction would tighten closure.

Cross-arc note: the M3 chain closure is the **first substantive substrate-graph closure of a load-bearing OPEN in the corpus producing a verdict upgrade** (M2 $\rightarrow$ M3 for this paper). Compared with the chirality cascade (load-bearing #2: closure attempts rejected; substrate is chirality-symmetric), the exponential-growth attack (load-bearing #1) closes positively via DCGT vacuum-energy translation. The asymmetry is substantively informative — **ED’s substrate ontology supports exponential-growth dynamics via standard QFT/cosmology inheritance through DCGT, but does not support chirality-discrimination via existing primitives.**

5 Audit Table

#	Step	Label	Notes
1	P02, P04, P07, P09, P11, P12, P13	P	Paper_087.
2	Upstream inheritance block	I	Paper_012, 072, 073, 089, 090, 093, ED_SC_4_9, SC-4.x, ED_CCC, ED_GW_00, ED_Baryogenesis.
3	P-Inflation-Identification (naming convention)	P	Definitional per Paper_095 §2.3.
4	Γ_{diff} and Γ_{prod} as substrate-graph rates	D-via-I	Paper_073 + Paper_012 + Paper_072.
5	Diffusion–production imbalance regime in post-SCBU ignition	D-via-I	Row 4 + Paper_ED_CCC §3.7.
6	Saddle-Hessian gradient-sign imbalance under diffusion-dominated regime	D-via-I	Paper_ED_SC_4_9 + row 5.
7	Outward V1 propagation of unbalanced-saddle pattern at substrate-c	D-via-I	Row 6 + Paper_089 + Paper_093.
8	Primary horizon-resolution via SCBU inheritance	D-via-I	Paper_ED_CCC §3.6 + SC-4.x.
9	Termination via Γ_{prod} recovery	D-via-I	Rows 4, 5, 7 + Paper_ED_CCC §3.7.
10	Inflaton-particle ontology absent at substrate level	A $\rightarrow$ position	Substrate-ontology Replaced framing.
11	Substrate-graph derivation that diffusion-dominated regime is naturally produced	OPEN	Fine-tuning relocation question.
12	Substrate-graph derivation that inflation’s horizon-resolution is non-redundant with SCBU	OPEN	Double-counting question.

#	Step	Label	Notes
13	Substrate-graph derivation that region-growth produces exponential (not linear) scale-growth	D-via-I (CLOSED via M3 chain)	§3.8. Steps A–C audit-accepted at approx-vacuum-energy level; D–E standard inheritance.
13a	Quantitative substrate-graph derivation of $\rho_{\text{eff}}^{\text{const}}$ and H value (OPEN-4)	OPEN	Not load-bearing for M3.
14	Substrate-graph derivation of regime duration (~ 60 e-folds equivalent)	OPEN	§3.5.
15	Substrate-graph derivation of smooth termination	OPEN	§3.5.
16	Substrate-graph derivation of qualitative power-law spectral character	OPEN	§3.6.
17	Substrate-graph derivation of primordial-fluctuation quantitative spectrum	OPEN	INHERITED from Planck.
18	Verdict M3 (form-IDENTIFIED + value-INHERITED) within DCGT A→regime	A→position	Per Paper_095 §3.3; upgraded from M2 following row 13 closure.

Zero pure-D rows. **Seven D-via-I rows** (4–9 + row 13 newly closed via M3 chain). One A→position substrate-ontology Replaced framing row (10). **Five remaining OPEN rows (11, 12, 14–17) plus 13a quantitative-consolidation OPEN** — none load-bearing for the M3 verdict.

6 Falsification Criteria

- **F1 (sharpest):** Detection of an inflaton-particle quantum — irreducible scalar-field excitation behaving as a particle of an inflationary potential, not as a substrate-regime dynamic. Refutes §3.7 substrate-ontology replacement (the only “Replaced” element, row 10).
- **F2:** Substrate-graph proof that no $\Gamma_{\text{diff}} > \Gamma_{\text{prod}}$ regime is structurally accessible in the post-SCBU ignition phase under any parameter consistent with the rest of the corpus. Refutes §3.3 and the entire substrate composition. **Framework-killing.**
- **F3:** Substrate-graph proof that substrate-c-bounded region-growth cannot produce a $\dot{a} \propto a$ mechanism under any V1 + V5 + DCGT parameter consistent with the rest of the corpus. Refutes row 13 directionally.
- **F4:** Observation of primordial non-Gaussianity, isocurvature modes, or spectral features requiring continuous-scalar-field origin inconsistent with substrate-graph V1-transported noise. Refutes §3.6 (audit row 16 closure direction).

7 Position Statement

This paper sits within the **substrate-ontology research-program lineage** (’t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

Cosmic inflation in ED is the structural-transition dynamics of the post-SCBU ignition phase (Paper_ED_CCC §3.7) — the transient substrate-regime where $\Gamma_{\text{diff}} > \Gamma_{\text{prod}}$ produces an unbalanced saddle-Hessian-signature pattern. The load-bearing **substrate-graph derivation of exponential scale-growth $\dot{a} \propto a$ closes via the M3 substrate-graph chain (§3.8)**

— established across seven exponential-growth memos. Substrate-c is preserved at substrate-graph level; exponential growth is a continuum-side coarse-grained-geometry effect via DCGT translating the saturation-regime constant substrate-action density to a vacuum-energy-like continuum stress-energy with $w \approx -1$. Same de-Sitter mechanism standard cosmology uses; substrate-side analog audit-accepted at approximate-level qualifications.

Remaining OPEN items (rows 11, 12, 14–17, 13a) are substrate-research frontier items — **none load-bearing for the M3 framing**. Row 13a (OPEN-4, quantitative $\rho_{\text{eff}}^{\text{const}}$ and H value from substrate parameters) is a quantitative consolidation deferred to SC-series follow-up. Horizon-resolution receives both SCBU inheritance (Paper_ED_CCC §3.6) and an inflation-side contribution; the “**no fine-tuning**” framing is replaced by “**fine-tuning relocated**” to substrate-graph regime parameters. **Inflaton-particle ontology genuinely “Replaced”** at substrate level. Other GR-side machinery (driver, mechanism, termination, expansion rate) is now D-via-I-substrate-graph-derivable via the M3 chain, not just vocabulary-Identified.

Verdict: M3 (form-IDENTIFIED + value-INHERITED) within DCGT A→regime hydrodynamic-window per Paper_095 — upgraded from M2 following audited closure of load-bearing row 13 via M3 substrate-graph chain. **First substantive substrate-graph closure of a load-bearing OPEN in the corpus producing a verdict upgrade.** Cosmology-arc placement: inflation supplies the dynamics of the post-SCBU ignition phase, not a separate cosmic mechanism; SCBU supplies the homogeneous initial state; the two arcs are complementary substrate-side readings of standard cosmology’s post-boundary content.

Comparison with chirality-cascade outcome (Paper_ED_Baryogenesis): load-bearing #2 (chirality $\mathbb{Z}_2$) closure attempts were rejected at audit; arc stays M2 with one postulate (substrate is chirality-symmetric). Load-bearing #1 (exponential growth, this paper) closes at D-via-I via the M3 chain; arc upgrades M2 → M3. The asymmetry is substantively informative — **ED’s substrate ontology supports exponential-growth dynamics through DCGT vacuum-energy translation but not chirality-discrimination via existing primitives.**